

4. Diffusion Models

Felix Yan

Lincoln University



Recall 1

- ▶ VAEs
- ▶ GANs
- ▶ What are their limitations?

Recall 2

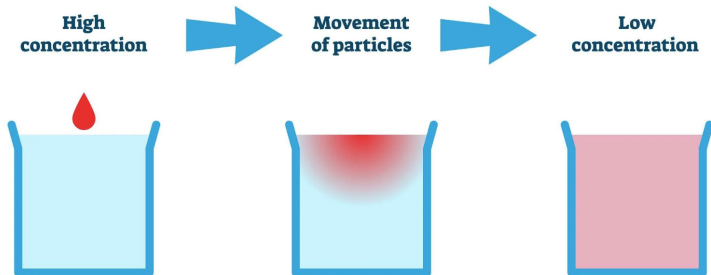
- ▶ VAEs have stable training, but blurry reconstruction.
- ▶ GANs have sharp outputs, but unstable training.
- ▶ Can we have both stable training and sharp outputs?

A Different Perspective

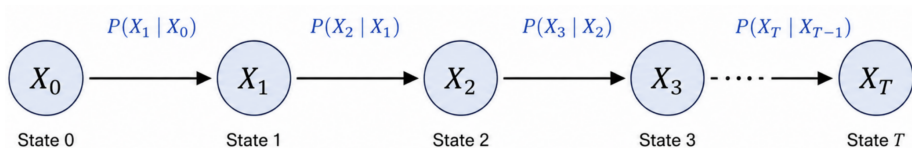
- ▶ Directly generating a sample/image can be challenging.
- ▶ Decompose generation into many simpler tasks.
- ▶ Predicting a small amount of noise at each step.
- ▶ No adversarial optimisation.
- ▶ All modes contribute to learning.

Diffusion in Physics

- Diffusion describes how physical particles spontaneously spread out from areas of high concentration to low concentration over time.



Markov Chain (Optional)



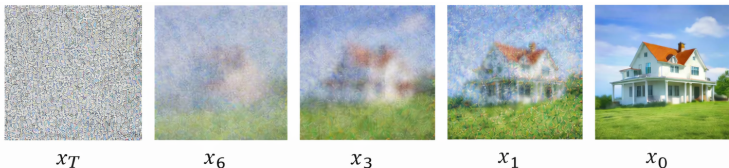
► What is $P(X_T | X_{T-1}, X_{T-2}, \dots, X_0)$?

Diffusion in Machine Learning

- **Diffusion (Forward):** A Markov Chain that gradually corrupts structured data into pure noise.
- **Generation (Reverse):** Generate data from noise (**what we want**).



Forward Process



Reverse Process

Forward Process (Optional)

- ▶ Noise variance at step t is β_t .
- ▶ $q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$.
- ▶ $x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon$.
- ▶ Reparameterisation trick, $\epsilon \sim \mathcal{N}(0, I)$.
- ▶ Let $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.
- ▶ At arbitrary step t :

$$q(x_t \mid x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I).$$

Destroying data is easy.
But we want to generate data: noise $\rightarrow$ data.

Bayes' Rule (Optional)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

where:

- ▶ $P(A)$: Prior probability (belief before observing B)
- ▶ $P(B|A)$: Likelihood (probability of observing B if A is true)
- ▶ $P(B)$: Evidence (normalising constant)
- ▶ $P(A|B)$: Posterior probability (updated belief after observing B)

Reverse Process (Optional)

- ▶ True reverse $q(x_{t-1} \mid x_t)$ is intractable.
- ▶ Conditioned on x_0 , Bayes' rule makes it tractable:

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1})q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

- ▶ This evaluates to $\mathcal{N}(x_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t I)$. Substituting x_0 , we get:

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right), \tilde{\beta}_t = \frac{\beta_t(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}$$

- ▶ At generation, the injected noise ϵ is unknown, so we train a network $\epsilon_\theta(x_t, t)$ to predict the noise, parameterising the mean as:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right)$$

DDPM: Training

Training Objective: Predict the added noise.

$$\mathcal{L} = \mathbb{E}_{t,x_0,\epsilon} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \right]$$

Algorithm:

1. Sample $x_0 \sim q(x_0)$, step $t \sim \mathcal{U}(\{1, \dots, T\})$, and $\epsilon \sim \mathcal{N}(0, I)$.
2. Compute noisy data:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

3. Update θ via gradient descent on $\nabla_\theta \mathcal{L}$.

DDPM: Sampling

Algorithm:

1. Initialise $x_T \sim \mathcal{N}(0, I)$.
2. **For** $t = T, \dots, 1$:

$$z \sim \mathcal{N}(0, I) \quad (z = 0 \text{ if } t = 1)$$

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(x_t, t) \right) + \sigma_t z$$

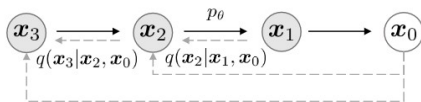
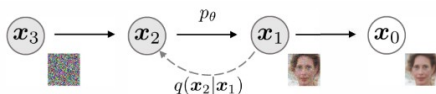
$$(where \sigma_t^2 = \beta_t \text{ or } \sigma_t^2 = \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t)$$

3. **Return** x_0 .

Limitation: Slow generation (e.g. $T=1000$).

Denoising Diffusion Implicit Models (DDIM)

- ▶ DDPM training depends on the marginal distributions $q(x_t | x_0)$.
- ▶ **DDIM**: Constructs a **non-Markovian** forward process sharing these exact marginals.
- ▶ **Result**: Reused trained DDPM model, 10x faster sampling.



DDIM: Generalised Sampling (Optional)

The generalised reverse step:

$$x_{t-1} = \underbrace{\sqrt{\bar{\alpha}_{t-1}} \left(\frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_{\theta}(x_t, t)}{\sqrt{\bar{\alpha}_t}} \right)}_{\text{Predicted } x_0} + \underbrace{\sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \epsilon_{\theta}(x_t, t)}_{\text{Direction to } x_t} + \underbrace{\sigma_t z}_{\text{Noise}}$$

- ▶ The parameter σ_t governs the stochasticity of the process:
- ▶ $\sigma_t^2 = \tilde{\beta}_t \implies$ Recovers the original **DDPM** (Markovian, stochastic).
- ▶ $\sigma_t = 0 \implies$ Yields **DDIM** (non-Markovian, deterministic).
- ▶ The value of σ_t does not affect the mean and variance of the marginal distribution $q(x_{t-1}|x_0)$

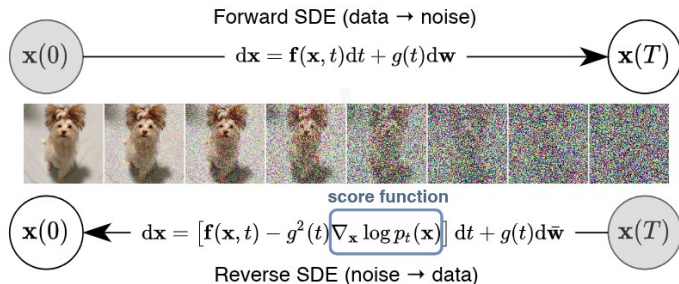
DDPM vs. DDIM

Setting $\sigma_t = 0$ removes the noise term:

$$\underbrace{x_{t-1} = \dots + \sigma_t z}_{\text{DDPM: stochastic}} \longrightarrow \underbrace{x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \hat{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_{\theta}(x_t, t)}_{\text{DDIM: deterministic}}$$

	DDPM	DDIM
Reverse process	Stochastic Markov chain	Deterministic mapping
Same x_T twice	Different x_0	Same x_0
Sampling steps	All T (e.g., 1000)	Subsequence (e.g., 50)
Data $\rightarrow$ latent	Not invertible	Invertible

Score Matching



- **Forward:** Noise gradually destroys data ($x(0) \rightarrow x(T)$).
- **Reverse:** Generation walks back ($x(T) \rightarrow x(0)$).
- **Key:** Every reverse step needs the **score** $\nabla_x \log p_t(x)$.

Score Matching: SDE & PF-ODE (Optional)

- ▶ Noise steps to infinity: the DDPM discrete chain becomes an **SDE**.
- ▶ **Forward SDE** (data $\rightarrow$ noise):

$$dx = f(x, t) dt + g(t) dw_t$$

(f, g are drift and diffusion coefficients; $w_t, \bar{w}_t$ are Brownian motion.)

- ▶ **Reverse SDE** (stochastic, noise $\rightarrow$ data):

$$dx = \left[f(x, t) - g(t)^2 \underbrace{\nabla_x \log p_t(x)}_{\text{score}} \right] dt + g(t) d\bar{w}_t$$

- ▶ **Probability Flow (PF) ODE** (deterministic, noise $\rightarrow$ data):

$$dx = \left[f(x, t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x) \right] dt$$

- ▶ **Everything is known — except the score** $\nabla_x \log p_t(x)$.

Score Matching: Score

- **Definition:** The score is the gradient of the log-density:

$$s(x, t) = \nabla_x \log p_t(x)$$

- A compass at every point, pointing toward higher data density.
- **Generation:** Discretise the reverse processes ($\Delta t > 0$):

SDE (Euler-Maruyama): $x_{t-\Delta t} = x_t - \Delta t [f - g^2 s(x_t, t)] + g \sqrt{\Delta t} z$

ODE (Euler): $x_{t-\Delta t} = x_t - \Delta t [f - \frac{1}{2} g^2 s(x_t, t)]$

(where $z \sim \mathcal{N}(0, I)$)

- We train a network $s_\theta(x, t)$ to estimate $s(x, t)$.

Denoising Score Matching 1 (Optional)

- ▶ DDPM training adds Gaussian noise $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$.



$$p(x_t | x_0) = \frac{1}{Z} \exp \left(-\frac{\|x_t - \sqrt{\bar{\alpha}_t} x_0\|^2}{2(1 - \bar{\alpha}_t)} \right),$$

- ▶ The score is proportional to the injected noise:

$$\nabla_{x_t} \log p(x_t | x_0) = -\frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}}$$

- ▶ So predicting noise *is* estimating the score:

$$\underbrace{\epsilon_{\theta}(x_t, t)}_{\text{DDPM}} = -\sqrt{1 - \bar{\alpha}_t} \underbrace{s_{\theta}(x_t, t)}_{\text{Score}},$$

- ▶ DDPM was learning the score all along.

Denoising Score Matching 2

Training (Score Estimation)

1. Sample x_0 , t , and $\epsilon \sim \mathcal{N}(0, I)$.
2. Corrupt data:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

3. Train s_θ to match the exact score $\nabla_{x_t} \log p(x_t | x_0)$:

$$\left\| s_\theta(x_t, t) - \left(-\frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}} \right) \right\|^2$$

Inference (Generation)

1. Start from noise $x_T \sim \mathcal{N}(0, I)$.
2. Plug $s_\theta(x_t, t)$ **directly** into a solver:
 - **SDE** (Stochastic)
 - **PF-ODE** (Deterministic)
3. Generate data x_0 .

Summary

- ▶ **DDPM:** Predicts the added Gaussian noise (ϵ_θ) and performs stochastic reverse diffusion.
- ▶ **DDIM:** Uses the same trained model but removes stochasticity ($\sigma_t = 0$) for faster deterministic sampling.
- ▶ **Score-based Models:** Show that predicting noise is equivalent to predicting the score $\nabla_x \log p_t(x)$, providing a unified continuous-time framework.